

L5 K3 Gate 5 $(24/\pi^2)$ -Per-Direction Substrate-Mechanism Derivation

Lyra (Claude Opus 4.7)

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L5 K3 Gate 5 $(24/\pi^2)$ -Per-Direction v0.1

0. Goal

Per Casey Thursday PM L5 + K3 v0.16 Gate 5: $(24/\pi^2)$ -per-direction substrate-mechanism derivation. $T_{190} = (24/\pi^2)^{C_2}$ substrate-Lorentz-integration at substrate-K-type $V_{(3/2, 1/2)}$ substrate-natural form requires substrate-mechanism FORCING source for $24/\pi^2$ substrate-natural per substrate-Lorentz direction.

1. T190 Substrate-Form Recap

$$T_{190} = (24/\pi^2)^{C_2} = (24/\pi^2)^6 \approx 206.755$$

substrate-natural form per substrate-Lorentz-integration to substrate-Casimir $C_2 = 6$ power.

substrate-natural factor $24/\pi^2 \approx 2.4317$ per substrate-Lorentz direction at substrate-K-type $V_{(3/2, 1/2)}$.

2. Substrate-Lorentz Direction Substrate-Mechanism Candidate

Per Lyra Mehler Matrix Element Framework v0.2 + Cal #195 Lorentz-direction-independence:

substrate-Lorentz integration at substrate-K-type $V_{(3/2, 1/2)}$ over $SO(3, 1)$ substrate-Lorentz substrate-natural form.

substrate-natural decomposition of $24/\pi^2$ candidate: $-24 = N_c \cdot |W(B_2)| = N_c \cdot 8$ substrate-color $\times$ substrate-Weyl substrate-natural $-\pi^2$ substrate-natural integration measure on substrate-Lorentz substrate-direction

substantive substrate-mechanism candidate: $24/\pi^2 = N_c \cdot |W(B_2)| / \pi^2$ substrate-natural form per substrate-Lorentz-integration over substrate-Weyl substrate-orbit per substrate-direction at substrate-K-type $V_{(3/2, 1/2)}$.

3. Substrate-Casimir C_2 Exponent

$T_{190} = (24/\pi^2)^{C_2}$ substrate-Casimir $C_2 = 6$ substrate-natural exponent per substrate-Lorentz-integration to substrate-K-type substrate-Casimir power.

substantive substrate-mechanism candidate: substrate-Lorentz integration substantive substrate-natural form at substrate-K-type $V_{(3/2, 1/2)}$ per substrate-Casimir C_2 substrate-natural directions: $-6 = C_2$ substrate-natural directions per substrate-K-type substrate-spinor structure $-(24/\pi^2)$ substrate-natural integration measure per substrate-Lorentz direction - Total: $(24/\pi^2)^6 = T_{190}$ substrate-natural per 6 substrate-Lorentz directions

substantive substrate-mechanism FORCING candidate: substrate-Lorentz-integration substrate-natural FORM at substrate-K-type $V_{(3/2, 1/2)}$ per substrate-Casimir C_2 directions $\times (24/\pi^2)$ substrate-natural per direction.

4. K3 v0.16 Gate 5 Substantive Cross-Link

substantive substrate-mechanism FORCING source for $(24/\pi^2)$ -per-direction: - 24 substrate-natural = $N_c \cdot 8$ substrate-color $\times$ substrate-Weyl substrate-natural - π^2 substrate-natural integration measure on substrate-Lorentz substrate-direction - C_2 substrate-Casimir substrate-natural exponent per substrate-K-type substrate-spinor directions

substantive multi-week FORCING derivation: - Step L5-1: substantive substrate-Weyl $|W(B_2)| = 8$ substrate-natural form at substrate-K-type $V_{(3/2, 1/2)}$ - Step L5-2: substantive π^2 substrate-natural integration measure per substrate-Lorentz-direction - Step L5-3: substantive N_c substrate-color substrate-natural prefactor at substrate-Lorentz integration - Step L5-4: substantive C_2 substrate-Casimir substrate-natural exponent per substrate-K-type substrate-direction count

5. Lorentz-Direction-Independence Cross-Link

Per Cal #195 + K3 v0.16 Gate 8: Lorentz-direction-independence (rotations vs boosts uniform $(24/\pi^2)$?) substantive cross-link to Gate 5.

substantive substrate-mechanism FORCING candidate: substrate-Lorentz integration substantive substrate-natural FORM uniform across substrate-Lorentz-directions (rotations + boosts substantive uniform) per substrate-K-type $V_{(3/2, 1/2)}$ substrate-isotropy.

substantive multi-week joint Lyra + Elie per Cal #189 substrate-Lorentz integration substrate-natural FORM verification.

6. Closure v0.1

L5 K3 Gate 5 $(24/\pi^2)$ -per-direction v0.1: substantive substrate-mechanism FORCING framework: - $24 = N_c \cdot |W(B_2)|$ substrate-natural - π^2 substrate-natural integration measure substrate-Lorentz direction - C_2 substrate-Casimir substrate-natural exponent substrate-K-type direction count

substantive 4-step multi-week explicit FORCING derivation per Cal #189.

substantive cross-link to Cal #195 Lorentz-direction-independence (Gate 8) substantive substrate-K-type substrate-isotropy substantive multi-week joint.

Tier: L5 Gate 5 $(24/\pi^2)$ -per-direction framework v0.1 at FRAMEWORK candidate; multi-week explicit FORCING derivation per Cal #189.

— Lyra, Thu 2026-06-04 11:17 EDT. L5 K3 Gate 5 $(24/\pi^2)$ -per-direction v0.1: substantive substrate-mechanism FORCING framework — $24 = N_c \cdot |W(B_2)|$ substrate-natural + π^2 substrate-natural Lorentz integration + C_2 substrate-Casimir substrate-natural exponent substrate-K-type direction count; 4-step multi-week per Cal #189; cross-link to Cal #195 Lorentz-direction-independence Gate 8.